AI for Medical Imaging

Master Artificial Intelligence

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2026/10/01 – Classification

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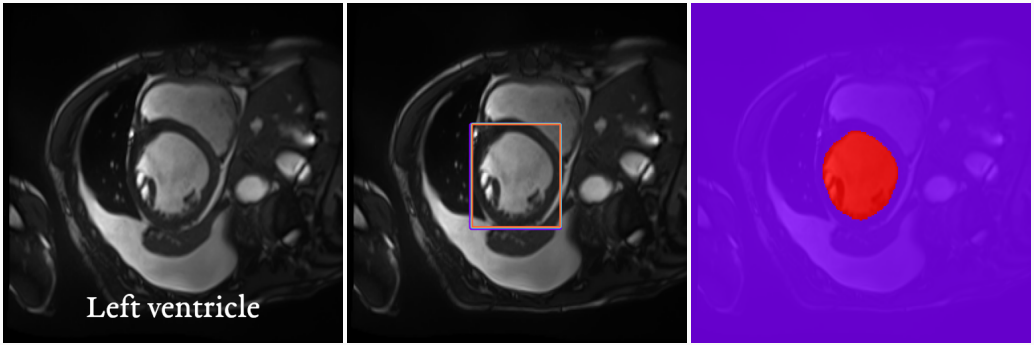
Announcements

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- Anki Deck almost finished
- Project will be 50% of the final grade (instead of 40%)
- Adding a thesis project page on the Canvas
- Likely shorter lecture today

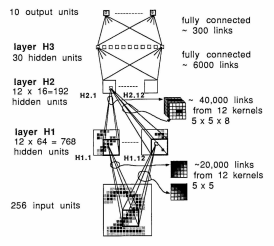
2026/10/01 – Classification

Classification?



Classification | Object detection | Image segmentation

Convolutional neural networks (again)



LeCun, Yann, et al. "Backpropagation applied to handwritten zip code recognition." Neural computation 1.4 (1989): 541-551.

Note: input and output are of fixed size.

Many new "state-of-the-art" architectures

Concrete example

Histopathology

What is the question, actually?

Will the patient survive? (Binary)

For how long? Binary? Bins/multi-class?

Regression?

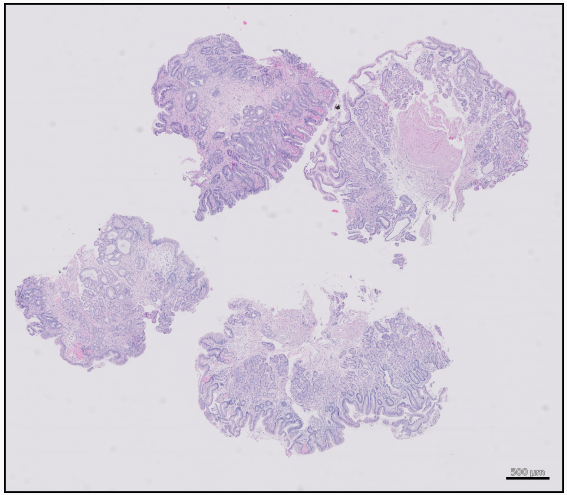
Is the label reliable?

“Everything in medical imaging is a maybe” — Me

2026/10/01 – Classification

Panels, consensus, agreements and disagreements

A case study: Barrett esophagus



4 possible labels: no dysplasia (NDBE), low-grade (LGD), high-grade (HGD, indefinite (IND)

Dutch Esophageal Pathology Panel (DEPP)

Reviews difficult cases, sent from the country: panel of experts, which evolves over time



Participation to the panel over time, by the 20 experts, over 10 years.

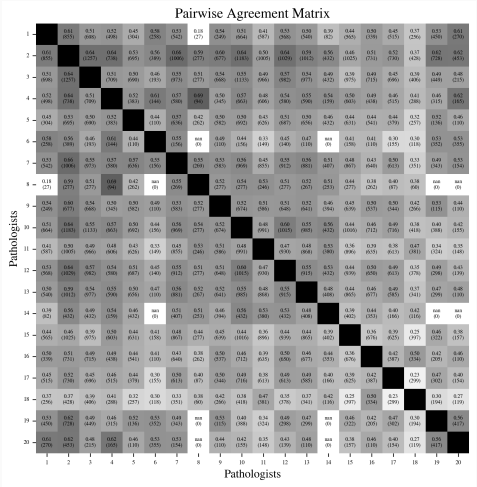
Multi-raters: what is a panel

Multiple pathologists meet at fixed time.

If more than 75% of the people in the room agree: consensus label

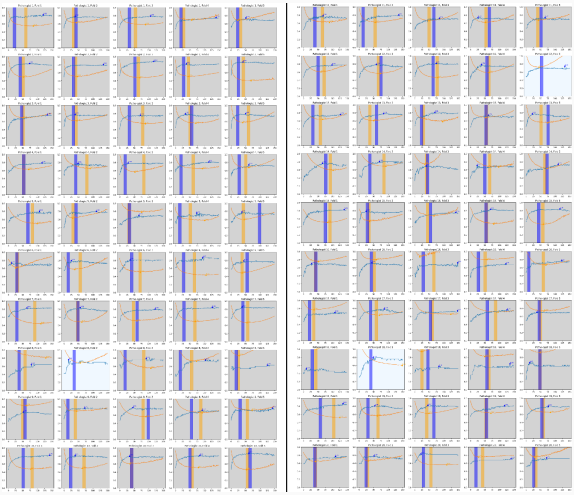
If not: *discuss until a consensus is reached*

Do they agree with each others?



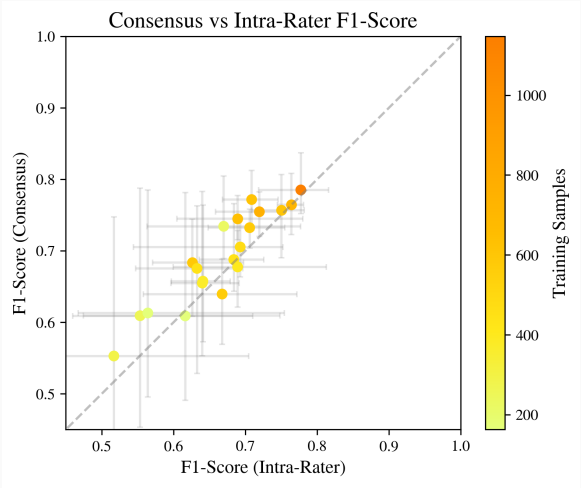
Training an AI: from which expert should you get your labels?

Idea: train one network per rater



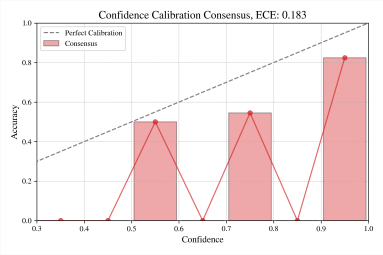
Most but not all networks converged

Intra vs consensus performances

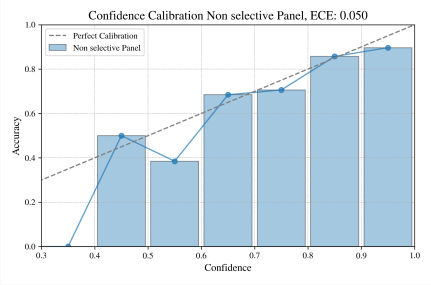


Somehow the intra-rater performance is correlated to the consensus performance.

Make a virtual panel: also interpretable and better calibrated



Single consensus network



Virtual panel

ECE for different settings

Better performances on this specific application, and free estimation of *uncertainty*

Take-away

- There are ways to leverage possibly conflicting labels
- Assessing uncertainty is as important as the prediction itself
- This is a whole field in itself

Confidence, uncertainty, and interpretability are all intertwined.

Great venue for the area of research: UNSURE workshop:
`https://unsuremiccai.github.io/`

Special issues at `https://www.melba-journal.org/issues/unsure25.html` and older



“Computer says no”

https://www.youtube.com/watch?v=0n_Ty_72Qds

2026/10/01 – Classification

Multiple-instance learning

The problem

Some images are too big, with a singe label for the whole

Information *somewhere*, but do not know where

Cannot downsample without losing information

Until recently: used Multiple Instance Learning (MIL)

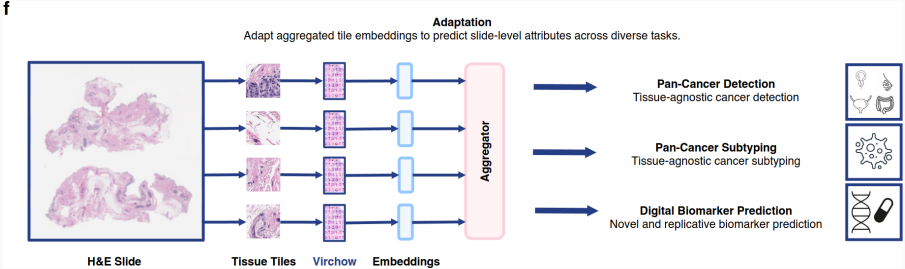
Idea of bags:

Positive bag: one sample (or more) has a positive label

Negative bag: no sample has a positive label

Sub-patched WSI (whole slide image) is a bag of patches

FOMO to the rescue



From *Virchow: A Million-Slide Digital Pathology Foundation Model*

Split into many patches, filtering out the background

Compute their embeddings with models such as Virchow

Learn from those (variable architectures)

The aggregator: Attention, self-attention, others

Several way to use the different Embeddings, different scenario:

1) Try to find the “smoking gun”: need to find only one patch (easier)

2) The combination and (spatial) relationship of patches makes the diagnostic (more difficult)

Very application dependent

2026/10/01 – Classification

What is the question, again? And concluding

Quantification vs *decision*

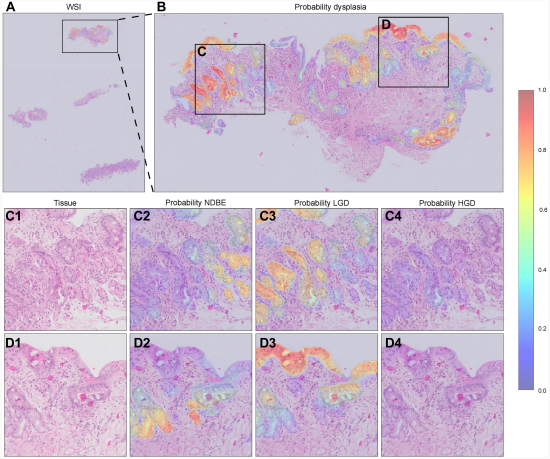
Discrete categories may be subjective, and ambiguous

Remember: we want to help the clinicians, not replace them.

Quantification, to provide better *biomarkers* for the clinicians to decide, is appealing.

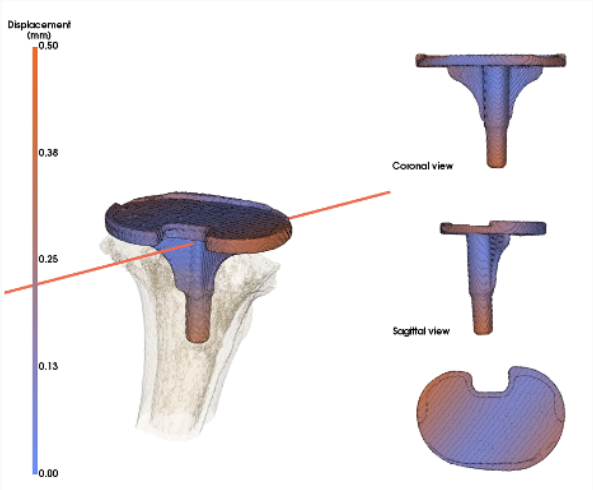
Also helps with interpretability.

Barret Esophagus: with a probability map?



From *Deep Learning for Histopathological Assessment of Esophageal Adenocarcinoma Precursor Lesions*, M. Botros et al., Modern Pathology, 2024,
https://doi.org/10.1016/j.modpat.2024.100531

Knee implant loosening?



AtMoves (guest talk next week): implant loosening detection

Take-aways

Many difficulties for classification:

- limited dataset;
- (very) big images;
- ambiguity of the labels

Take the time to really think: what is the question worth asking